

Appendix E - Brief User Guide

This appendix guides the reader through installing PHAROS and creating, running, and inspecting their own scenarios in the graphical application or with the standalone runner. The public repository provides the Windows release, source code, and technical documentation (Raiher, 2026b). Detailed descriptions of scenario files, command-line execution, and the controller interface are supplied in `Docs/Reference/TGSCN.md`, `Docs/GettingStarted/StandaloneRunner.md`, and `ControllerSDK/PHAROSControllerAPI.h`, respectively.

The screenshots in this appendix are unaltered full-frame captures of the English-only PHAROS interface. They are visual landmarks rather than an exhaustive image catalog: the tables and text describe options that may lie below the visible scroll position. A later software release may differ from these captures and should be used with its own bundled documentation.

E.1 Installation, Data, and a Portable Mission Folder

PHAROS is a 64-bit Windows desktop application. Run the Windows installer and open PHAROS from the Start Menu or the optional desktop shortcut. The default installation directory is `%LOCALAPPDATA%\Programs\PHAROS`; `PHAROS.exe` can also be launched directly from that directory. Neither Unreal Engine nor Visual Studio is required to use the installed application or its standalone runner. On first launch, the application asks the user to review and accept its end-user license agreement.

User data are stored separately from the installation, below `%LOCALAPPDATA%\PHAROS\Saved`. This directory contains the scenario library, controller workspaces, simulation results, settings, and diagnostic logs. Routine library backups are retained under `SaveGames\Backups`; an unreadable or unsupported library is copied to `SaveGames\Recovery` before replacement is attempted.

Launching PHAROS opens the main menu shown in Figure E.1. *New Scenario* begins a blank authoring workflow, *Scenario Library* opens saved definitions and retained results, *About & Documentation* gives access to the bundled reference material, and *Exit* closes the application.



FIGURE E.1 – PHAROS main menu and its four primary entry points: new-scenario authoring, the Scenario Library, documentation, and application exit.

A shareable simulation should be organized as a self-contained mission folder, for example:

```
mission/
  mission.tgscn
  controllers/
  gravity/
  atmosphere/
  aerodynamics/
  propulsion/
  geometry/
  textures/
  results/
```

All referenced files may use absolute or TGSCN-relative paths. Relative paths are resolved from the directory containing the TGSCN file and should be preferred for material intended for another computer. Record the origin, units, validity range, and license of each physical or visual input. A custom controller DLL is native executable code, is not sandboxed, and should be loaded only when its source and provenance are trusted.

E.2 Scenario Authoring

The configuration screen can create a new scenario, import a portable TGSCN, save the current draft, or open the Scenario Library. A robust authoring order is to define the time and numerical method, enter the initial state, build the spacecraft component tree, add actuators and

control, select only the required environment models, and finish in the Review panel. Table E.1 summarizes the principal settings in each panel.

TABLE E.1 – Configuration panels and principal options in PHAROS.

Panel	Principal Options and Responsibility
Scenario and Solver	Scenario name; initial UTC; end by duration or final UTC; fixed-step RK4 or adaptive Dormand–Prince 5(4); step, absolute-tolerance, and relative-tolerance settings; output at every completed integration interval or at a fixed recording interval; numerical and stored-output safety limits.
Initial State	Input in ICRF or a supported body’s SPICE body-fixed frame at the initial epoch; Cartesian position and velocity; scalar-first attitude quaternion; angular velocity; displayed frame, normalization, and unit conventions. The portable Physics Core state is canonical barycentric ICRF in SI units.
Component Tree	One rooted hierarchy of rigid components; component mass, minimum/variable mass, local center of mass, centroidal inertia, parent and child anchors, zero orientation, and ordered rotational or translational degrees of freedom; joint coordinates, rates, limits, and effort limits; primitive or imported-STL visual geometry and material appearance.
Actuators	Prescribed-profile or controller-commanded thrusters; mount and variable-mass propellant references; application point and direction; ignition/shutdown window; thrust and specific impulse; fixed-axis reaction or momentum wheels, initial momentum, and saturation limit.
Controls	No controller or one compiled user C++ controller; selection from the managed library; compatibility with Controller API version 3. The library can create source from the template, import source, build it with the bundled toolchain, or import an advanced prebuilt DLL after an explicit trust decision.
Gravity and Bodies	Supported Solar System members and barycenters; gravity enablement; automatic activation and barycenter-resolution radii; first post-Newtonian correction; optional harmonic-gravity CSV and maximum degree. Required SPICE data are checked before execution.
Solar Radiation Pressure	Global enablement; celestial eclipses; mutual shadowing by articulated spacecraft components; default absorption, specular, and diffuse fractions; component inclusion, surface tessellation, and optional component-specific optical properties.
Atmosphere	Central body and available atmospheric data source; profile or model inputs and their validity preview. The selected density, thermodynamic, and molecular/collision data feed the aerodynamic model rather than the renderer.
Aerodynamics	Reference area and length; minimum dynamic pressure for evaluating loads and maximum dynamic pressure for the validity diagnostic; constant-drag fallback and/or a scattered coefficient database; interpolation/extrapolation policy; molecular-speed-ratio, Knudsen-number, flow-direction, and optional articulation-coordinate dimensions; force and moment coefficients.
Review	Aggregated syntax, reference, physical-consistency, data-file, controller, SPICE, and run-readiness diagnostics. Errors block acceptance. Warnings require explicit acknowledgement, and acceptance applies only to the exact reviewed draft revision.

Figures E.2–E.4 illustrate the numerical, state, and spacecraft-definition pages. The central component preview is a construction aid; only physical properties, topology, joint definitions, and explicitly enabled physical surfaces enter the Physics Core. Meshes, textures, colors, and other presentation fields survive TGSCN round-trip but do not alter the propagated equations.

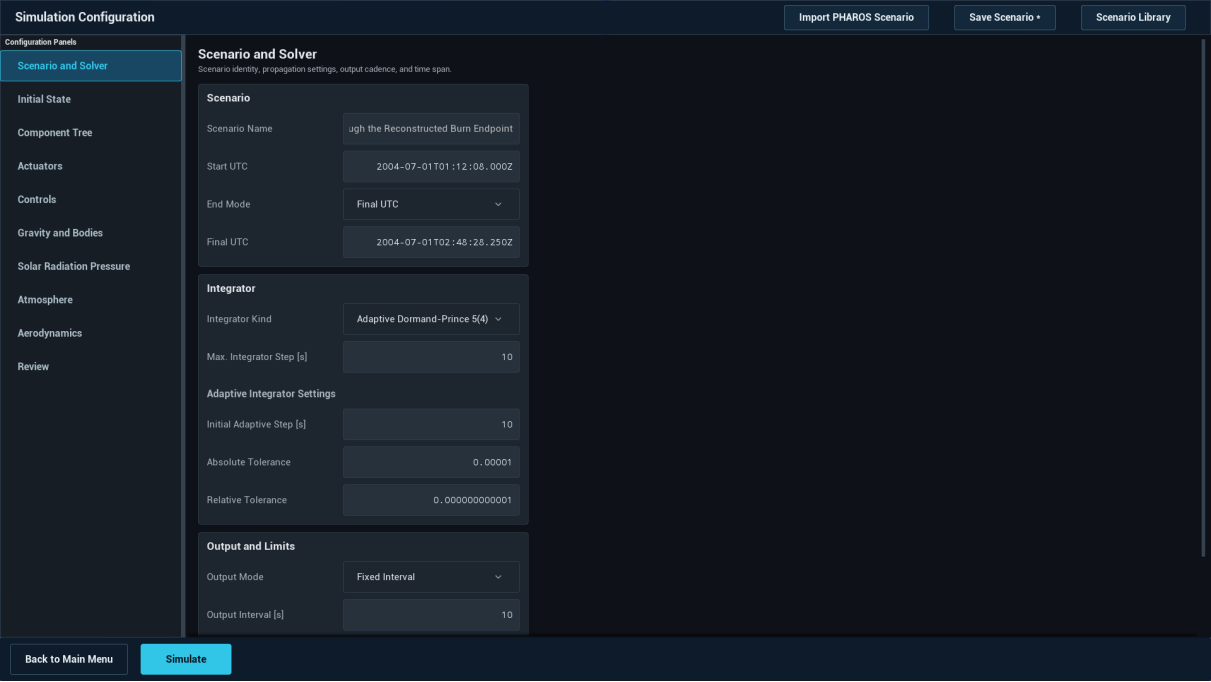


FIGURE E.2 – Scenario duration, numerical integrator, output cadence, and safety limits in the Scenario and Solver panel.

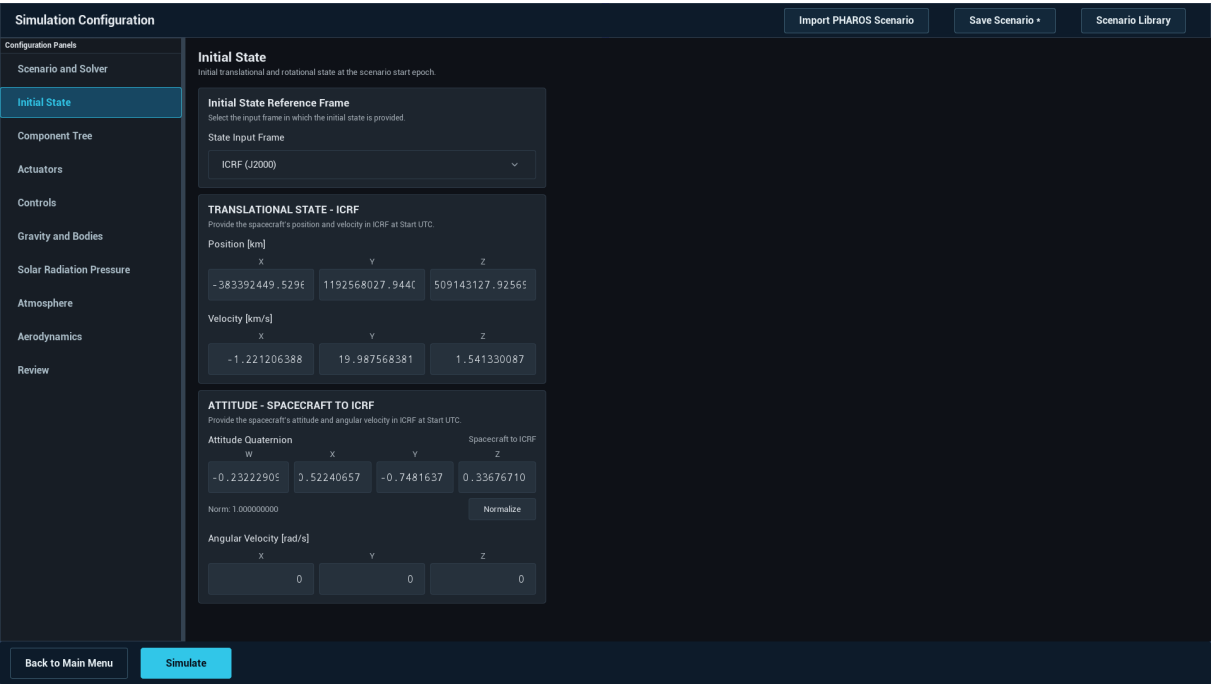


FIGURE E.3 – Reference-frame selection and translational and rotational initial-state entry.

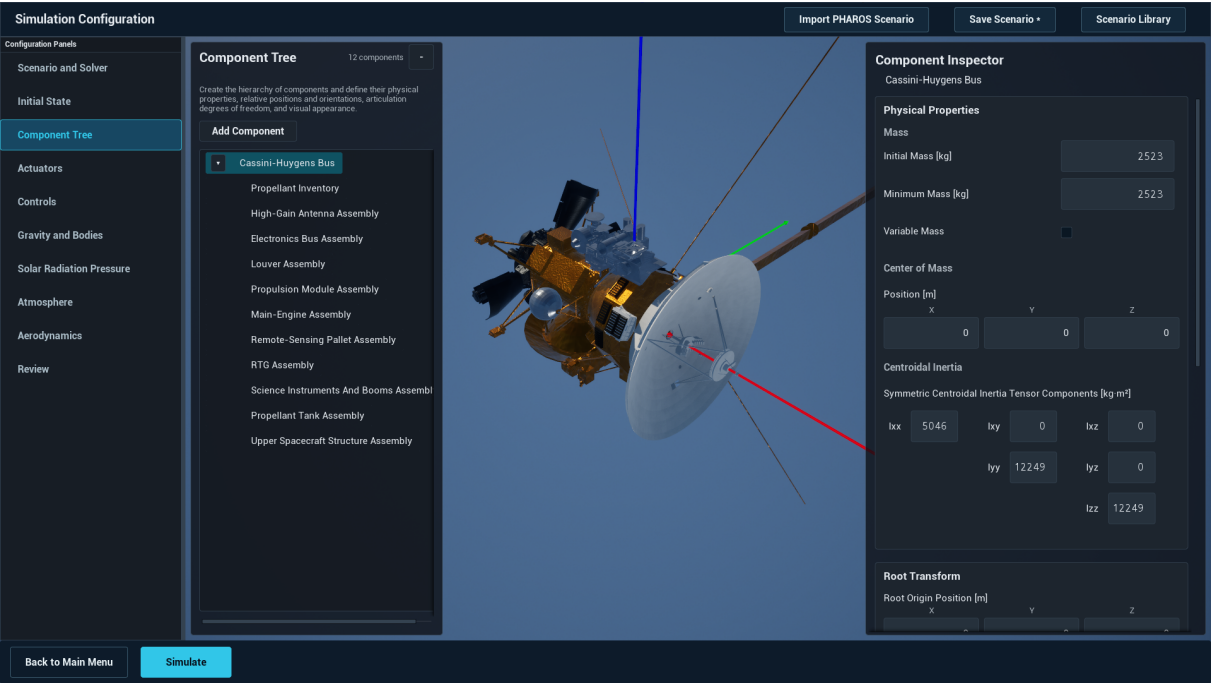


FIGURE E.4 – Hierarchical component tree, three-dimensional preview, and selected-component physical-property inspector.

The actuator pages in Figures E.5 and E.6 do not create hidden hardware properties. The mass and inertia of an engine or wheel must be represented by the component tree; the actuator record supplies the force, torque, mass-flow, or stored-momentum interaction. Every thruster must be mounted on a component and must reference a variable-mass propellant component.

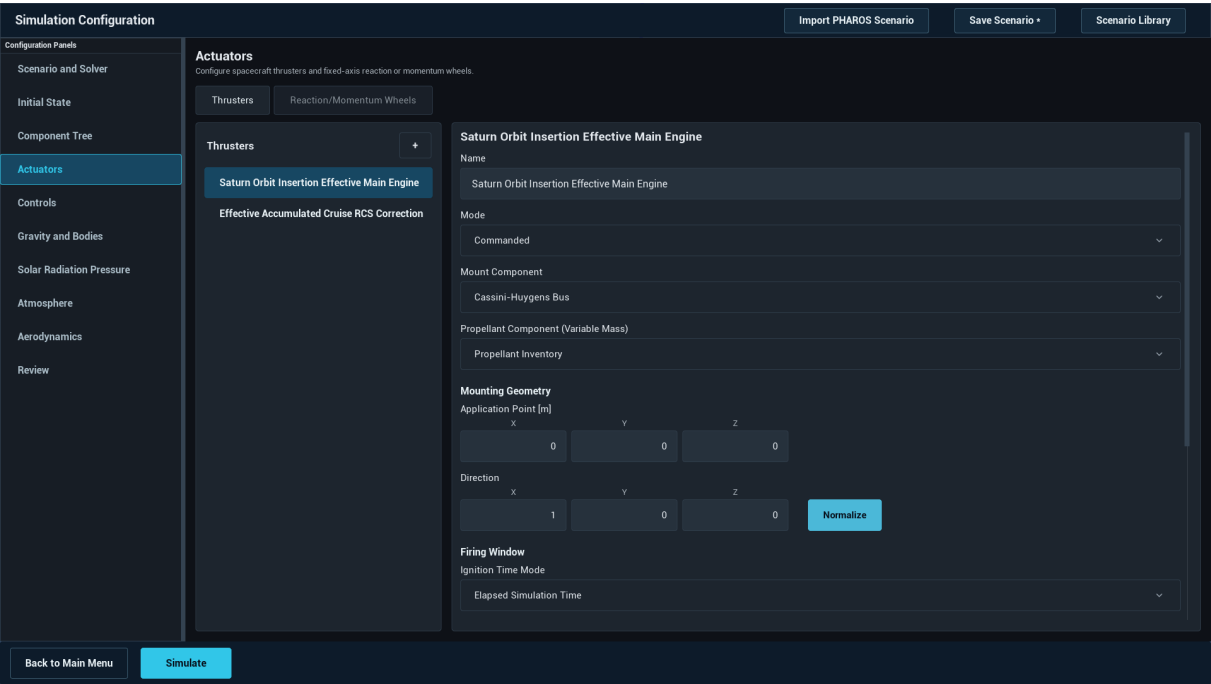


FIGURE E.5 – Thruster list and the commanded-thruster mounting, propellant, direction, and firing-window fields.

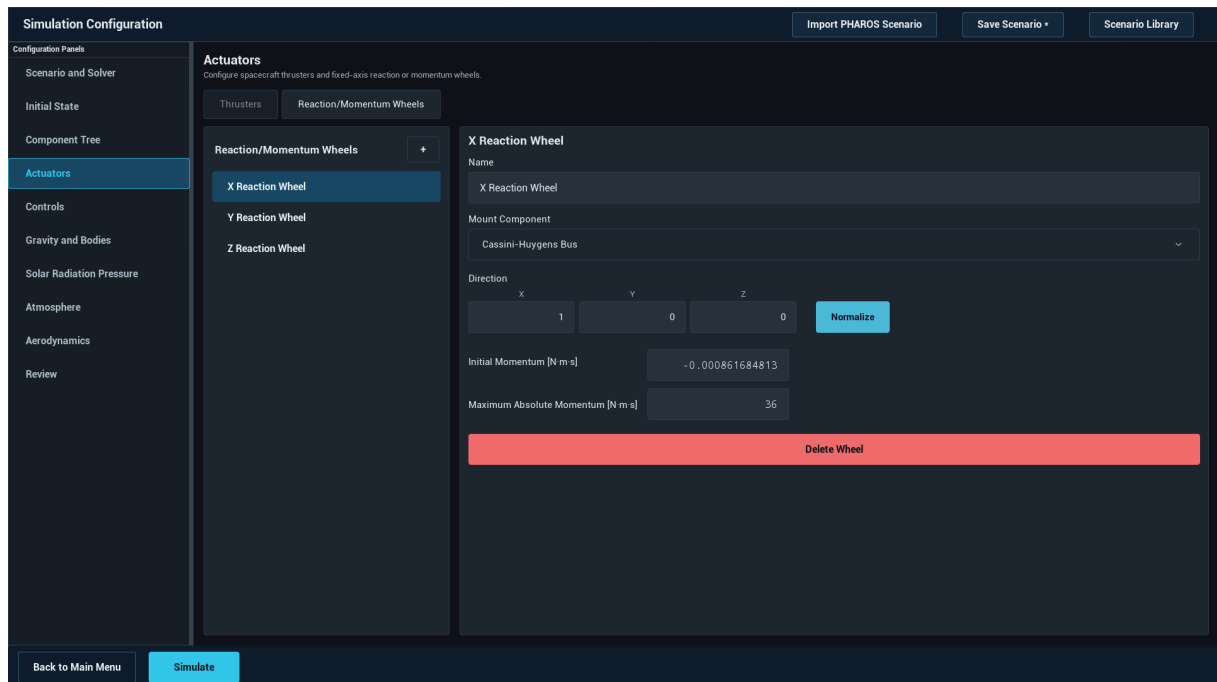


FIGURE E.6 – Fixed-axis reaction/momentum-wheel definition, including its mount, axis, initial momentum, and saturation limit.

E.2.1 User Controllers

Selecting *User C++ Controller* associates one ready Controller Library entry with the scenario, as illustrated in Figures E.7 and E.8. The library’s template and reference header define Controller API version 3. At each requested evaluation, PHAROS gives the controller read-only time, spacecraft, component, joint, actuator, celestial-body, and environment data. The controller returns throttle and specific impulse for commanded thrusters, joint efforts, wheel-momentum rates, and an optional direct external body torque. It may also supply discharge derivatives and the time of its next command discontinuity, as specified in Appendix D. It cannot overwrite the propagated position, velocity, attitude, angular velocity, mass, joint state, or wheel momentum.

Identify components and actuators by the names supplied in the API views rather than assuming an array order. Command buffers are reset before each evaluation, so omitted actuator commands remain zero; optional discharge derivatives remain unspecified unless explicitly supplied. Nonfinite commands are rejected, and the actuator models apply the configured operating limits. An incompatible DLL must be rebuilt against the supplied API before it can be used. Appendix D provides the template and the rules for repeated controller evaluations.

E.2.2 Environment and External Loads

Figures E.9–E.12 show the four environment pages. Enable only models justified by the intended experiment: additional options increase the required data and assumptions and do not by themselves increase fidelity. Imported harmonic, atmosphere, and aerodynamic files

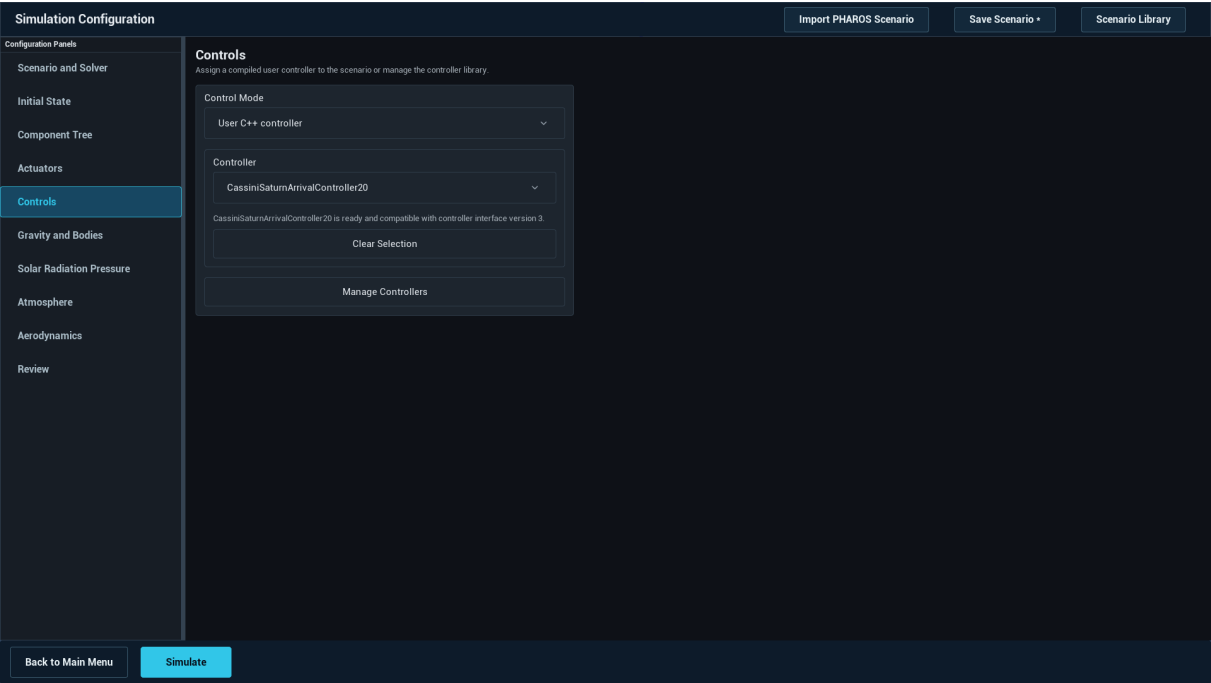


FIGURE E.7 – Scenario control mode and selection of a ready, API-compatible managed controller.

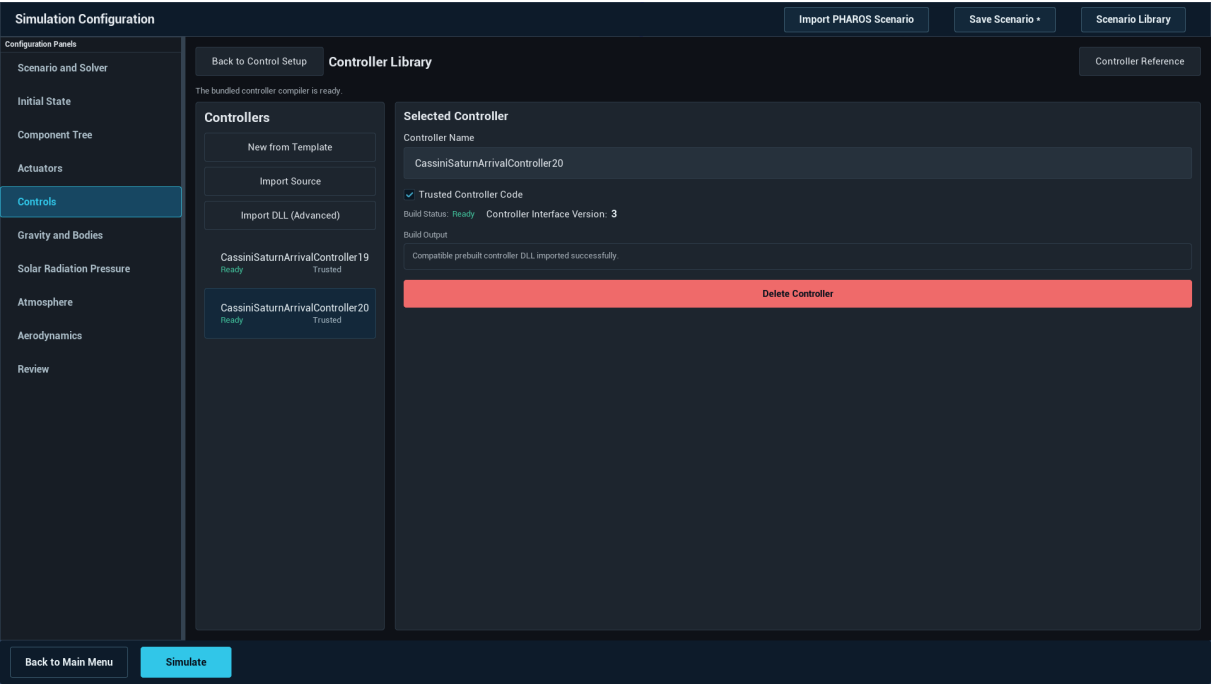


FIGURE E.8 – Controller Library with source-template, source-import, advanced DLL-import, trust, build-status, and interface-version controls.

are previewed and checked before a run. Visual illumination in Unreal is independent of the numerical SRP, eclipse, atmospheric, and aerodynamic calculations.

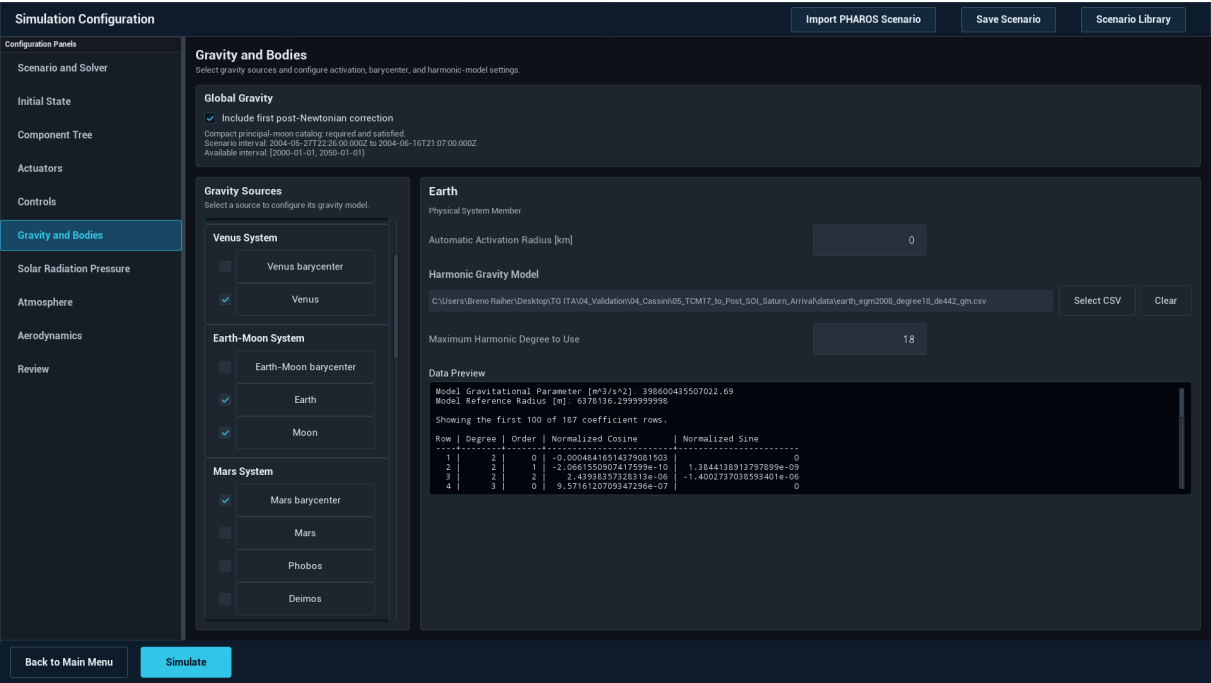


FIGURE E.9 – Gravity-body catalog, relativistic correction, activation policy, and an imported harmonic-gravity model.

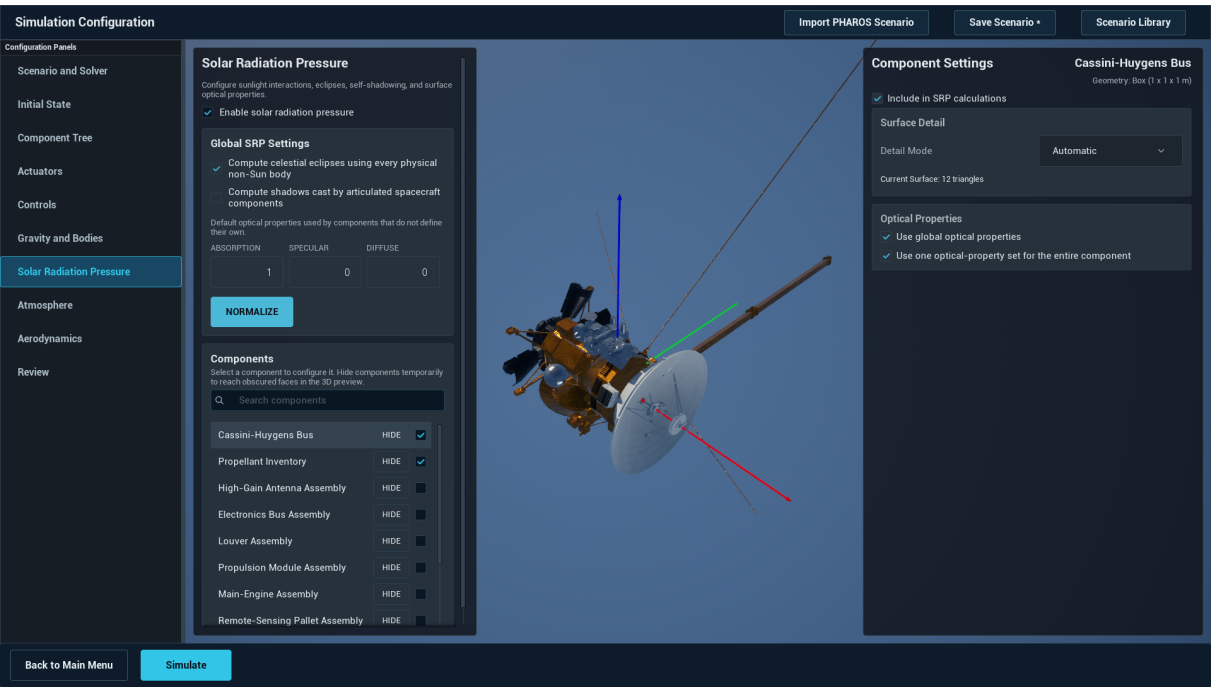


FIGURE E.10 – Solar-radiation-pressure configuration with eclipse, spacecraft-shadowing, surface-detail, and optical-property controls.

E.3 Review, Simulation, and the Scenario Library

Right-click *Simulate* and select *Limit Execution Time*. This supervisor limit stops the runner after the specified real execution time in seconds, independently of the simulated duration.

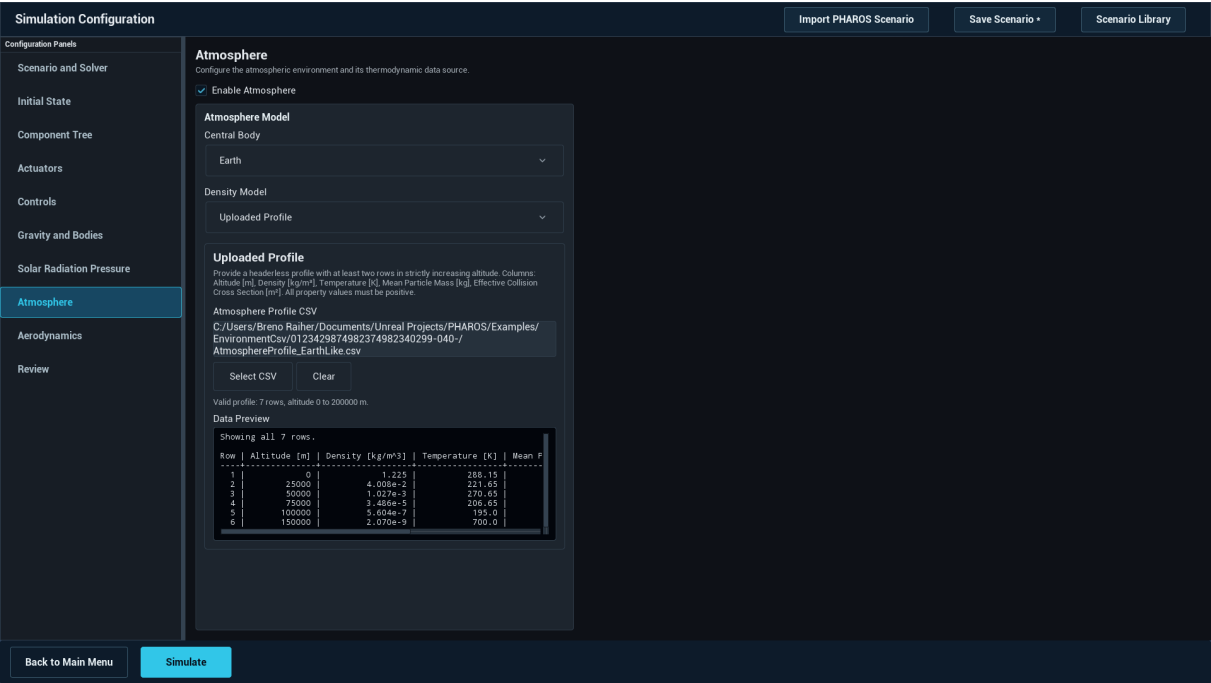


FIGURE E.11 – Atmosphere selection and validation preview for an uploaded altitude-dependent profile.

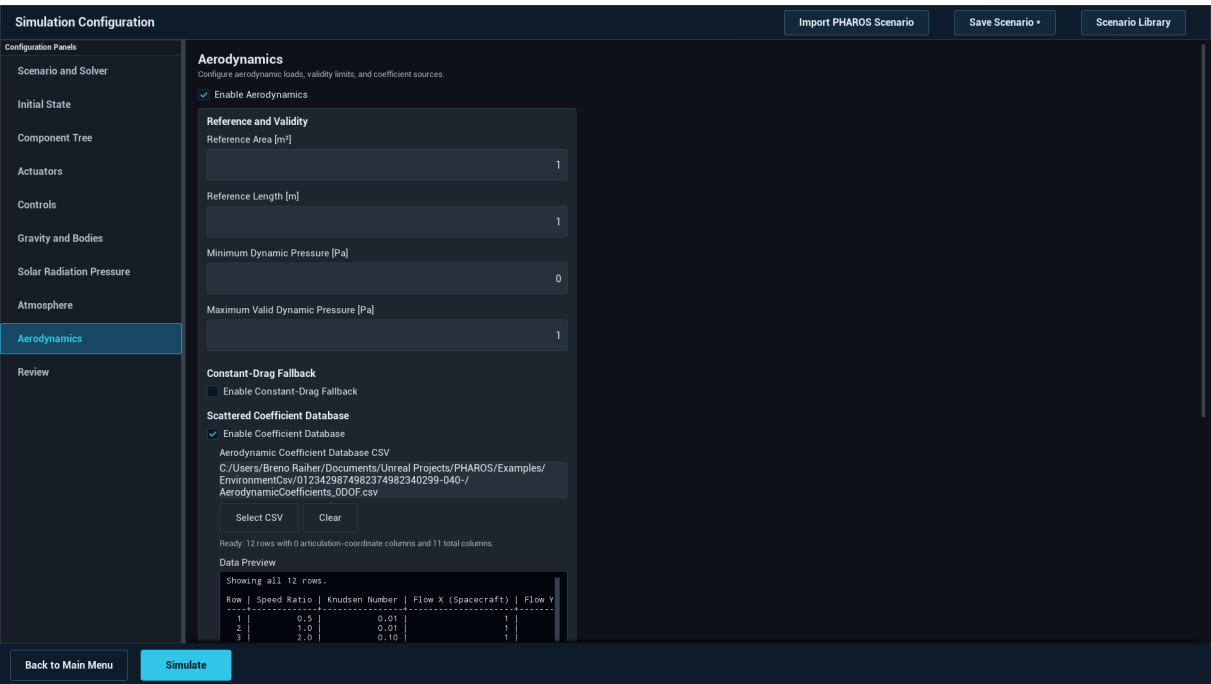


FIGURE E.12 – Aerodynamic validity limits and coefficient-source configuration, with a tabulated-data preview.

The Review panel must validate the exact draft revision before the *Simulate* command is enabled. Selecting an issue navigates back to the responsible configuration field. Errors cannot be accepted. Warnings may be accepted only after the user acknowledges them, as in Figure E.13; any later edit invalidates that acceptance and requires a new review.

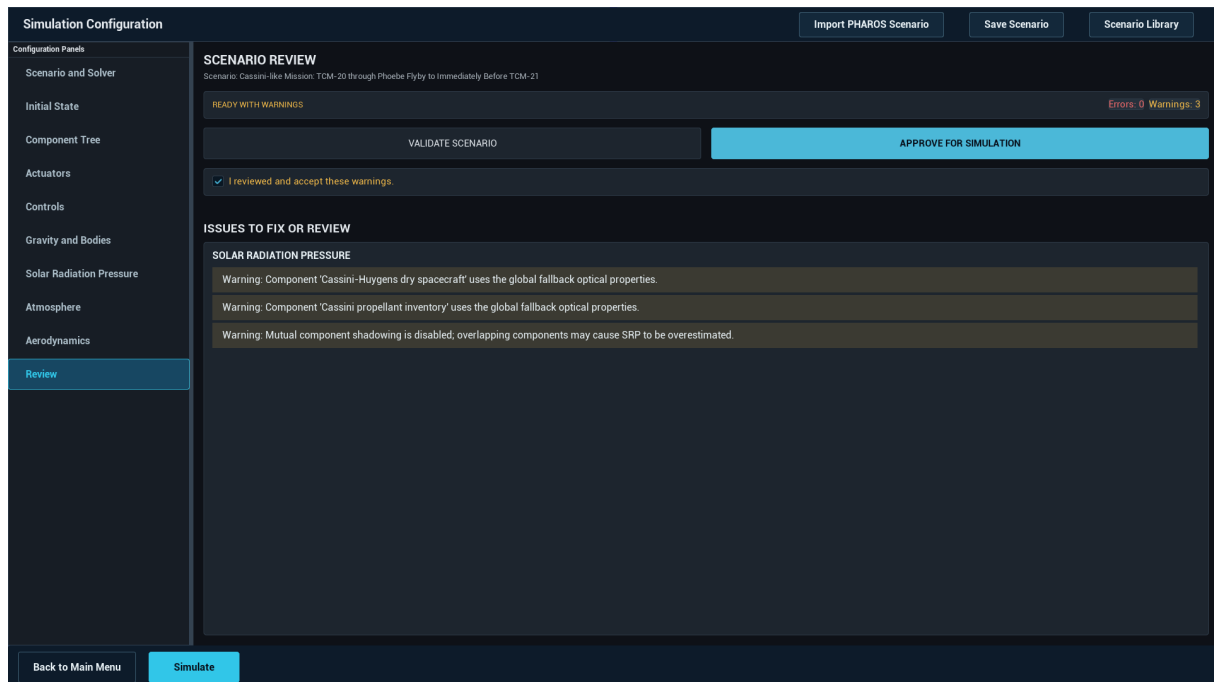


FIGURE E.13 – Approved Review report with no errors and three explicitly acknowledged SRP warnings.

For an application-run scenario, PHAROS exports a TGSCN snapshot and starts `PHAROSScenarioRunner.exe` as a separate process. The application displays progress and diagnostics, accepts cancellation, and enforces the execution-time limit. A completed run stores the scenario snapshot, solution CSV, summary, and log; its status and file paths are registered in the Scenario Library after the result is checked. The TGSCN and CSV must remain available for continuation or playback. Important results should be copied out of the application data directory together with their dependent files.

The Scenario Library provides saved-scenario and completed-run ownership. It can reopen a scenario for editing, load a completed result for visualization, or continue from a retained terminal state. A continuation is a new scenario whose initial dynamic state is derived from the selected result; topology-changing events such as staging or separation still have to be represented outside one continuous Physics Core run.

Figure E.14 shows a populated library. The search, sorting, refresh, recent-items, and complete-list controls support retrieval as the number of scenarios grows. Each row exposes whether a result package is available together with creation and modification times. After a compatible row is selected, the context-sensitive toolbar can reopen the scenario, open its visualization, create a continuation from the propagated terminal state, or delete the library entry.

E.4 Standalone Validation and Propagation

The standalone runner uses the same scenario preparation, physical models, and recorder as the graphical application, without opening Unreal. The following PowerShell examples use

SCENARIO LIBRARY

Open Scenario Open Visualization Continue from End of Propagation Delete Refresh

Q Search scenarios Sort by Last Modified Newest first

RECENT

Scenario	Result Status	Last Modified	Created
Earth orbit - attitude-thruster Sun-pointing recovery (visual-appearance variant)	Result available	Sep 4, 2026, 12:45 AM	Sep 4, 2026, 12:45 AM
Earth Orbit - Four-Wheel Sun-Pointing Recovery	Result available	Sep 4, 2026, 12:35 AM	Sep 4, 2026, 12:35 AM
JWST CONTINUOUS MCC1B_BURN	Result available	Sep 4, 2026, 12:12 AM	Sep 4, 2026, 12:12 AM

ALL SCENARIOS 15 scenarios (0 selected)

Scenario	Result Status	Last Modified	Created
Earth orbit - attitude-thruster Sun-pointing recovery (visual-appearance variant)	Result available	Sep 4, 2026, 12:45 AM	Sep 4, 2026, 12:45 AM
Earth Orbit - Four-Wheel Sun-Pointing Recovery	Result available	Sep 4, 2026, 12:35 AM	Sep 4, 2026, 12:35 AM
JWST CONTINUOUS POST_INSERTION_L2_COAST	No result	Sep 4, 2026, 12:23 AM	Sep 4, 2026, 12:23 AM
JWST CONTINUOUS MCC1B_BURN	Result available	Sep 4, 2026, 12:12 AM	Sep 4, 2026, 12:12 AM
JWST CONTINUOUS COAST_TO_MCC1A	Result available	Sep 3, 2026, 11:57 PM	Sep 3, 2026, 11:57 PM
Apollo 8-like Mission: Transearth Injection Controlled Maneuver	Result available	Sep 3, 2026, 10:30 PM	Sep 3, 2026, 10:30 PM
Apollo 8-like Mission: Approach to Lunar Orbit Insertion Attitude Start	Result available	Sep 3, 2026, 10:20 PM	Sep 3, 2026, 10:20 PM
Apollo 8-like Mission: Post-Separation Coast to First Midcourse Correction Ignition	Result available	Sep 3, 2026, 10:11 PM	Sep 3, 2026, 10:11 PM
Cassini-like Mission: Finite TCM-14 through Jupiter Flyby to Immediately Before TCM-17	Result available	Sep 3, 2026, 9:22 PM	Sep 3, 2026, 9:22 PM
Cassini-like Mission: Finite TCM-7 through Venus-2 to Immediately Before TCM-9	Result available	Sep 3, 2026, 9:08 PM	Sep 3, 2026, 9:08 PM
Cassini-like Mission: Finite TCM-2 through Venus-1 to Immediately Before TCM-5	Result available	Sep 3, 2026, 8:31 PM	Sep 3, 2026, 8:31 PM

Back

FIGURE E.14 – Scenario Library with recent and complete scenario lists, result-availability status, search and ordering controls, and the context-sensitive commands for editing, visualization, continuation, and deletion.

the default installation directory; change `$pharosRoot` if another directory was selected. Run them from a mission folder containing `mission.tgscn` and its referenced resources. The SPICE kernels are separate data files supplied with PHAROS, not embedded in the executable. First validate the inputs without propagating:

```
$pharosRoot = "$env:LOCALAPPDATA\Programs\PHAROS"
$runner = "$pharosRoot\PHAROS\Binaries\Win64\PHAROSScenarioRunner.exe"
$kernel = "$pharosRoot\PHAROS\Content\SPICEKernels"
& $runner ".\mission.tgscn" --validate-only --kernel-dir $kernel
```

Then propagate to a separate output directory:

```
& $runner ".\mission.tgscn" `
  --output ".\results\run_001" --kernel-dir $kernel
```

Validation loads the kernels and any selected controller, reads the scenario, resolves its references, prepares the numerical inputs, and validates the Physics Core configuration. It does not integrate or write a solution CSV. A direct command-line invocation does not enforce the scenario's supervisor wall-clock limit; an external process manager is required when that deadline is mandatory.

The runner prints progress and diagnostics to the terminal and returns one of the codes in Table E.2. For `mission.tgscn`, a propagated run writes `mission_solution.csv` and `mission_summary.txt`. The CSV begins with absolute ephemeris time and independent

TABLE E.2 – Exit codes returned by PHAROS Scenario Runner.

Code	Meaning
0	Validation or propagation succeeded
2	Invalid command-line arguments
3	TGSCN parse failure
4	Controller DLL load or compatibility failure
5	SPICE-kernel load failure
6	Scenario preparation or configuration-validation failure
7	Output directory creation failure
8	Result-file write failure
9	Numerical propagation returned a failed result

elapsed time, followed by the recorded spacecraft, component, actuator, environment, force, and moment telemetry selected by the result recorder. A file's existence is not proof of success: inspect the exit code and summary before using any partial diagnostic output.

E.5 Result Visualization

Opening a completed run presents a spacecraft-centered three-dimensional view driven by the stored samples. The timeline can play, pause, change rate, and seek by mission time, ephemeris time, or UTC. Dockable panels provide:

- spacecraft, component, actuator, force, moment, and closest-body telemetry;
- heliocentric and closest-body two-dimensional trajectory overviews;
- selectable celestial-direction, state, force, and moment vectors;
- constellation outlines for orientation context; and
- two- or three-dimensional plots of compatible recorded columns over a selected time range.

Figures E.15–E.17 show representative panel arrangements. Refocus and Solar Overview change the view, not the dynamics. Spacecraft geometry is reconstructed from the TGSCN, while motion and telemetry come from the associated CSV. Rendering converts meters to Unreal centimeters and uses a moving visual origin; neither operation feeds back into the Physics Core.

Playback interpolates between recorded samples. A coarse output interval can hide a short event or make a straight display chord cross a body even when the integrated trajectory does not. Use the recorded state values and a suitable physical or numerical reference to assess the motion; the visualization helps inspect it.

E.6 First Inspection and Reproducibility

For every completed calculation:

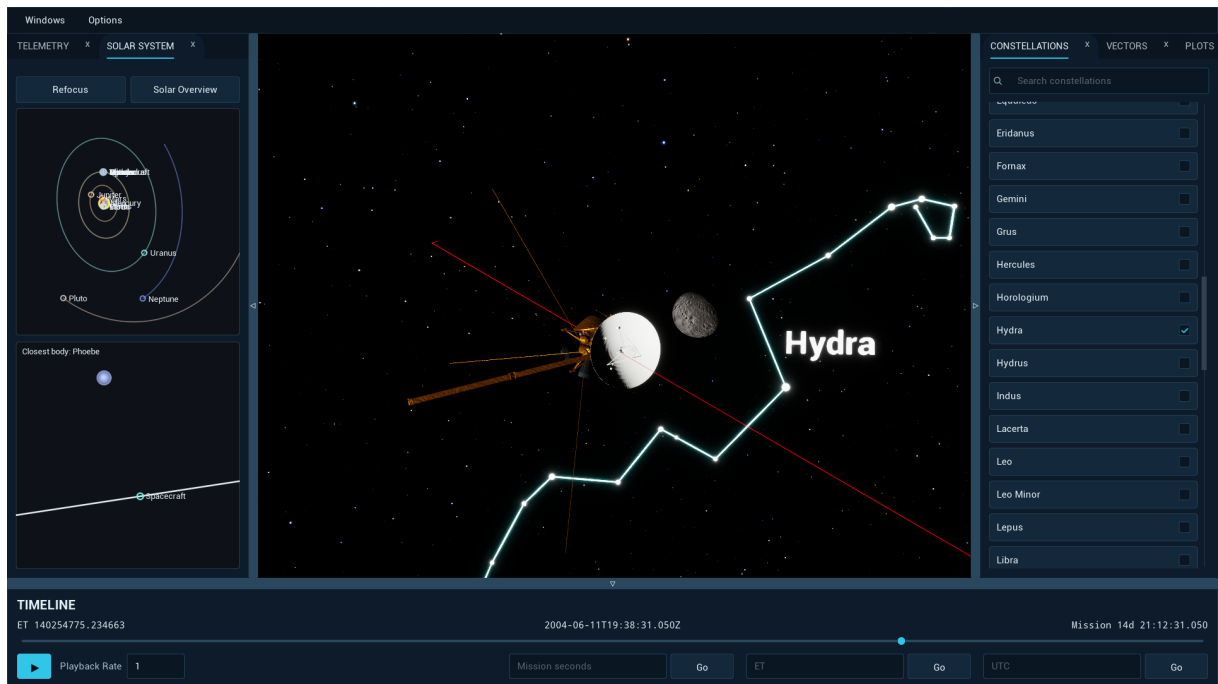


FIGURE E.15 – Playback with Solar System overviews and a selectable constellation outline.



FIGURE E.16 – Playback with spacecraft telemetry and an interactive plot built from recorded CSV columns.

1. confirm the process exit code, summary success flag/message, and expected final epoch;
2. verify sample and column counts against the configured output policy and safety limits;
3. scan for nonfinite values, unexpected discontinuities, and states outside configured limits;
4. check quaternion norm, joint limits, wheel saturation, variable-mass bounds, and propellant monotonicity during firing;
5. confirm the sign, frame, application point, and magnitude of forces and moments;

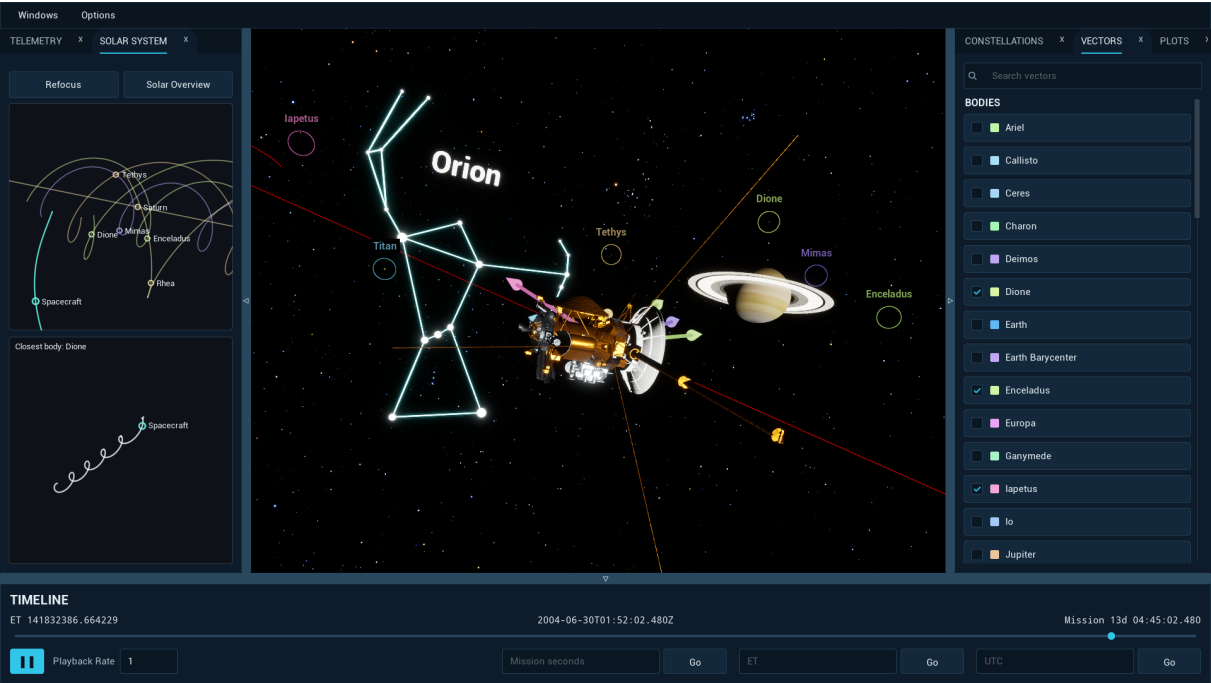


FIGURE E.17 – Close spacecraft view with the Saturn-system overview, constellation context, and selectable celestial vectors.

6. compare an invariant, analytical limit, independent implementation, or external truth source appropriate to the case; and
7. interpret the result only within Appendix A.

Table E.3 connects common diagnostic symptoms to the checks the user should make before continuing.

TABLE E.3 – Common failure categories and first checks.

Symptom	First Checks
Parse or Schema Failure	File marker/version, TOML brackets and quotes, finite values, unknown keys, and required tables
Missing External File	TGSCN-relative path, case/spelling, package completeness, permissions, and whether the file was moved
Time or Celestial-Body Failure	Kernel directory and load set, UTC spelling, catalog entry, selected body, and epoch coverage
Component or Joint Failure	Unique IDs/names, exactly one root, parent existence, cycles, mass and inertia, anchors, normalized axes/quaternions, and ordered DOFs
Propulsion Failure	Mount and variable-mass propellant references, thrust direction, I_{sp} , firing window, profile endpoints, throttle, and remaining propellant
Controller Failure	Explicit trust, API version 3, required exports, x64 architecture, build output, finite commands, and actuator name/index mapping
Integrator Stops Early	Reported failure, step/output limits, tolerance scale, nonfinite derivative, discontinuity timing, and force/profile jumps
Implausible Force or Moment	Units, resolving frame, application point, reference point, sign convention, duplicated model activation, and input validity range

Continued on the next page.

TABLE E.3 – Common failure categories and first checks (continued).

Symptom	First Checks
Misleading Playback	TGSCN–CSV pairing, output cadence, interpolation across short events, visual scale/origin, selected vectors, and camera reference

Preserve the TGSCN and its external inputs, controller source and DLL, kernel set, PHAROS version, runner hash, execution command, exit code, and original output files. Store analysis scripts and derived plots separately. Reproducible defects can be reported through the project repository (Raiher, 2026b).